

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination

May 2014

CL 102 Environmental Sciences

Date: 16.05.2014, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q-1 (a) Define following Term: [03]
1) Environment 2) Ecology 3) Ecosystem.
- Q-1 (b) Discuss relationship between Man and Environment. [04]
- Q-2 (a) Explain objective & guiding principles of Environmental Studies. [04]
- Q-2 (b) Define Noise Pollution. Describe Sources of Noise pollution. [05]
- Q-2 (c) What is drinking water quality standard? Why water quality standard needed? [05]
Describe Hardness parameter.

OR

- Q-2 (a) Describe Structure of Atmosphere in detail. [04]
- Q-2 (b) Explain the term "Environmental Degradation" and discuss the role of advanced Technology in the degradation of environment. [05]
- Q-2 (c) Describe effects of Air Pollution on plant And human being. [05]
- Q-3 (a) Explain the Sources & Effects of any four common air pollutants. [04]
- Q-3 (b) Define Environmental Pollution. Also classify various pollutants. [05]
- Q-3 (c) What is 'Acid Rain'? Write a brief note on effects of acid rain on environment. [05]

OR

- Q-3 (a) Enlist the common air pollutants stating their permissible limits as per ambient air Quality standards. [04]
- Q-3 (b) Enlist all current global environmental problems and Write a note on Global Warming. [05]
- Q-3 (c) Define Eutrophication, its causes and effects. [05]

SECTION – II

- Q-4 (a) Define an eco-system. Classify components of eco-systems. [03]
- Q-4 (b) Describe Energy flow in the eco-system with neat sketch. [04]
- Q-5 (a) Explain different types of ecological pyramids with their relative advantages & disadvantages. [05]
- Q-5 (b) (i) Explain the term 'Bio-magnification' with example. [03]
(ii) Explain food web in eco-system. [03]
- Q-5 (c) List out population forecasting methods & explain in detail Geometrical Increase method. [03]

OR

- Q-5 (a) Analyze aquarium as an ecosystem & enlist various biotic & abiotic components in it. [05]
- Q-5 (b) (i) The study of organism & its environment is called _____. [06]
(ii) Autotrophs mean those organisms who _____.
(iii) Examples of edaphic components are _____.
(iv) The food chain starting from dead organic matter is known as _____.
(v) Sulphur & phosphorous cycles are known as _____.
(vi) The energy flow in ecosystem is _____.
- Q-5 (c) Explain the methods for control of population growth. [03]
- Q-6 (a) Explain renewable & non renewable resources with examples. [03]
- Q-6 (b) Write short note on overuse of water. What are the problems associated with over exploitation of water resources? [04]
- Q-6 (c) Comment on urban energy problems & discuss the effect of overpopulation over energy problems in India. [07]

OR

- Q-6 (a) Write short note on destruction v/s conservation. [03]
- Q-6 (b) Write a note on geothermal & tidal energy. [04]
- Q-6 (c) Explain in detail causes of deforestation. Also mention its effects. [07]